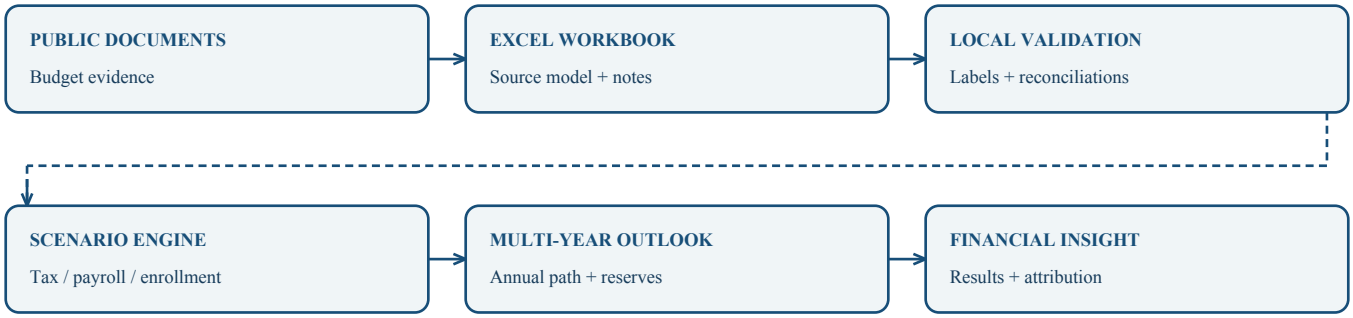
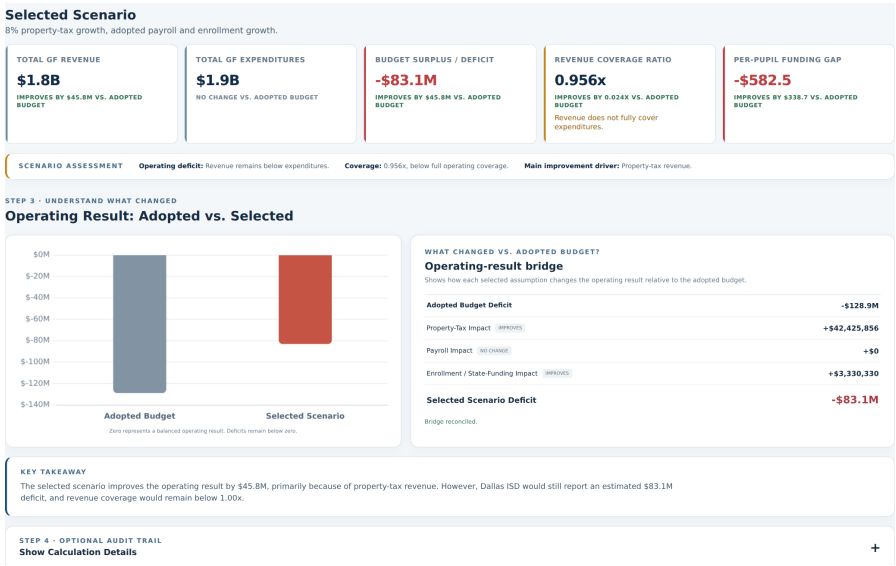
INDEPENDENT PUBLIC FINANCE RESEARCH

Dallas ISD Financial Risk & Scenario Analysis

Financial Research & Technical Methodology Report

From a public budget to an interactive investigation of fiscal resilience.





Current dashboard capture • illustrative revenue-upside scenario

Roberto Almeida

October 2026 • Final edition 1.0

FY2026 adopted budget baseline. Conditional scenarios; no rating or investment recommendation.

REPORT GUIDE

Contents

A research question, a traceable source model and an interactive analytical workflow. Sections move from the idea to the evidence, calculations and financial interpretation.

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How to read the results

Budget observations describe source planning amounts. Scenario results are calculated sensitivities. Multi-year outcomes depend on a user-defined annual path. None is presented as a realized outcome or rating forecast.

01 / THE PROBLEM AND THE IDEA

A budget is the starting point. What happens when conditions change?

This project began with research into the creditworthiness of U.S. public-sector issuers. That investigation led to school districts and then to Dallas Independent School District as a case study.

The research question

How financially resilient is Dallas ISD, and how could changing financial and operating conditions affect its long-term fiscal sustainability and credit profile?

Public budget documents provided the starting evidence. Organizing selected information in Excel made it possible to compare figures and calculate ratios. The next question was how changes in revenue, spending and enrollment would affect that position.

The Dallas ISD dashboard is an interactive web application. It imports the source workbook, validates required inputs and independently calculates the consequences of property-tax, payroll and enrollment assumptions. It also explains the change from the adopted budget and explores a future annual path.

What this project demonstrates

• Public financial research and structured Excel preparation.

• Required-input checks and financial reconciliations.

• Combined fiscal scenarios, sensitivity and mathematical break-even analysis.

• Reconciled risk attribution and multi-year reserve/shortfall analysis.

Who this is for

Analysts and researchers exploring budget sensitivity and financial flexibility; and students and learners practicing public-budget interpretation through explicit assumptions. Adapting the application to another issuer requires reviewing its mapping and methodology.

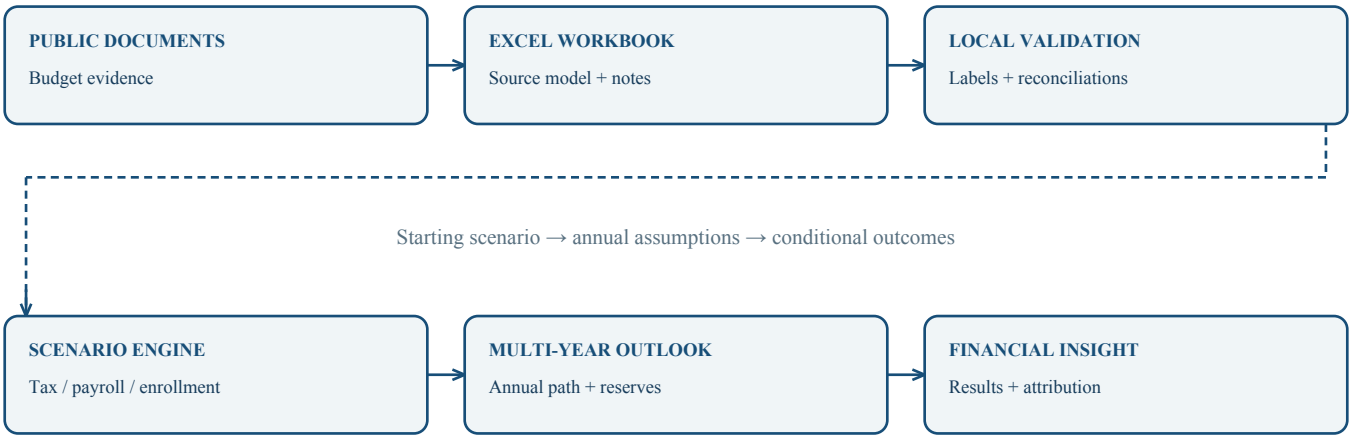
Core principle

Every modeled result should be traceable to source inputs, explicit assumptions and a documented calculation. The user should understand both what changed and why.

02 / ARCHITECTURE AT A GLANCE

One connected analytical workflow

Public documents supply evidence; Excel organizes the financial record; the application translates explicit assumptions into conditional results. The workbook is the source model, while the dashboard performs independent calculations.



Layer	Role
Source preparation	Selected budget figures, notes and reference calculations in seven Excel sheets.
Local intake	SheetJS parses the file in browser memory; labels and headers identify required inputs.
Financial checks	Revenue components and expenditure totals are reconciled before interpretation.
Shared scenario engine	Baseline, guided scenarios, attribution and break-even calculations.
Multi-year engine	Starting scenario plus annual compounding; recovery and reserve reconciliation.
Presentation and delivery	JavaScript interface, Chart.js visualizations and a single-file production build.

Scenario Analysis defines the selected starting scenario. Multi-Year Outlook applies future changes without changing those scenario selections. No backend, database or external financial-data API is used.

03 / INSIDE THE DASHBOARD

Establish the starting position

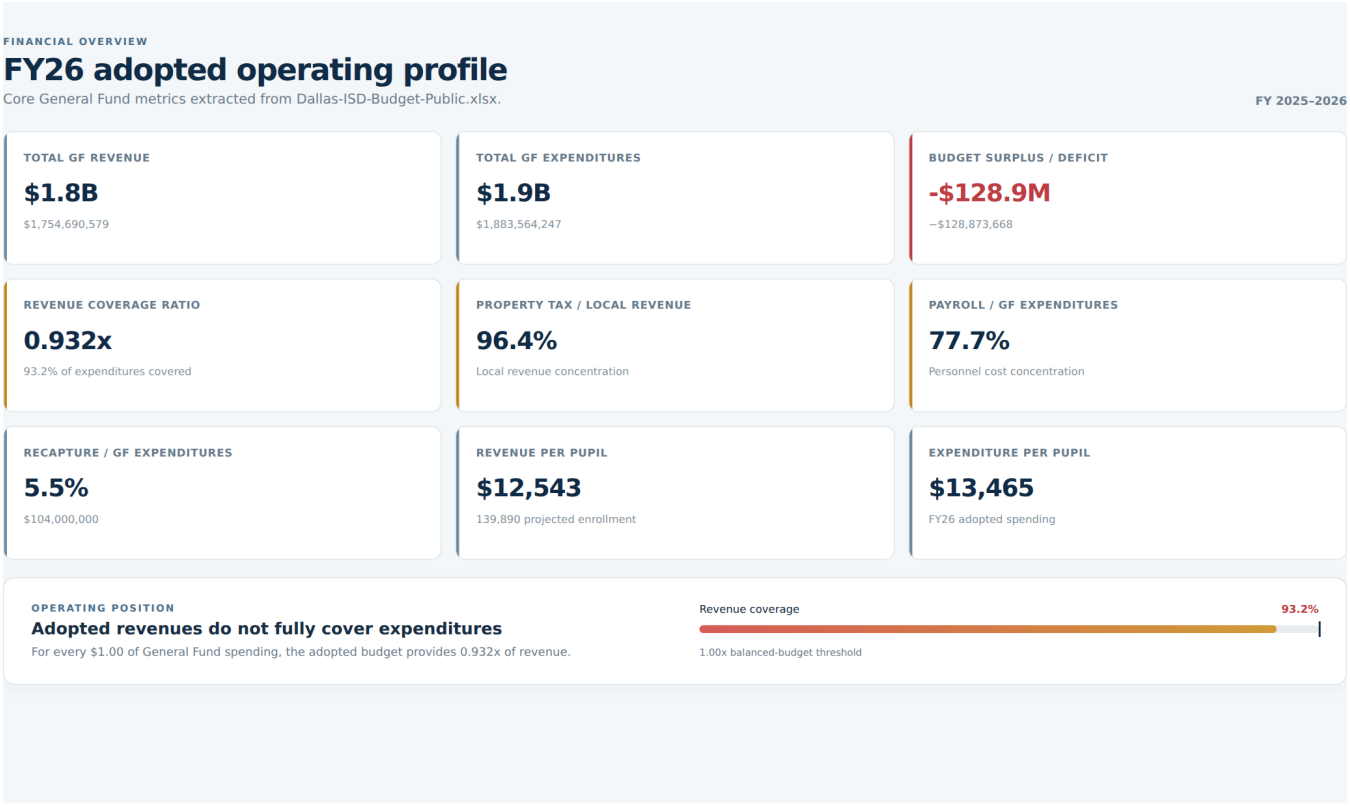


Figure 1. Actual Financial Overview capture from the standalone application; adopted budget inputs.

FY2026 General Fund revenue of $1.755B is below expenditures of $1.884B. The resulting budgeted deficit is $128.9M and coverage is 0.932x. These are planning figures, not actual financial results. [1, pp. 25, 31, 35]

Read the baseline before choosing a scenario

The overview provides operating results, financial composition and per-pupil context. Property taxes are approximately 96.4% of local revenue; payroll is approximately 77.7% of total General Fund expenditures. These denominators must remain explicit.

What the screen answers

Where does the imported budget stand, and which revenue and spending categories are large enough to make their assumptions material?

03 / INSIDE THE DASHBOARD

Choose explicit fiscal assumptions

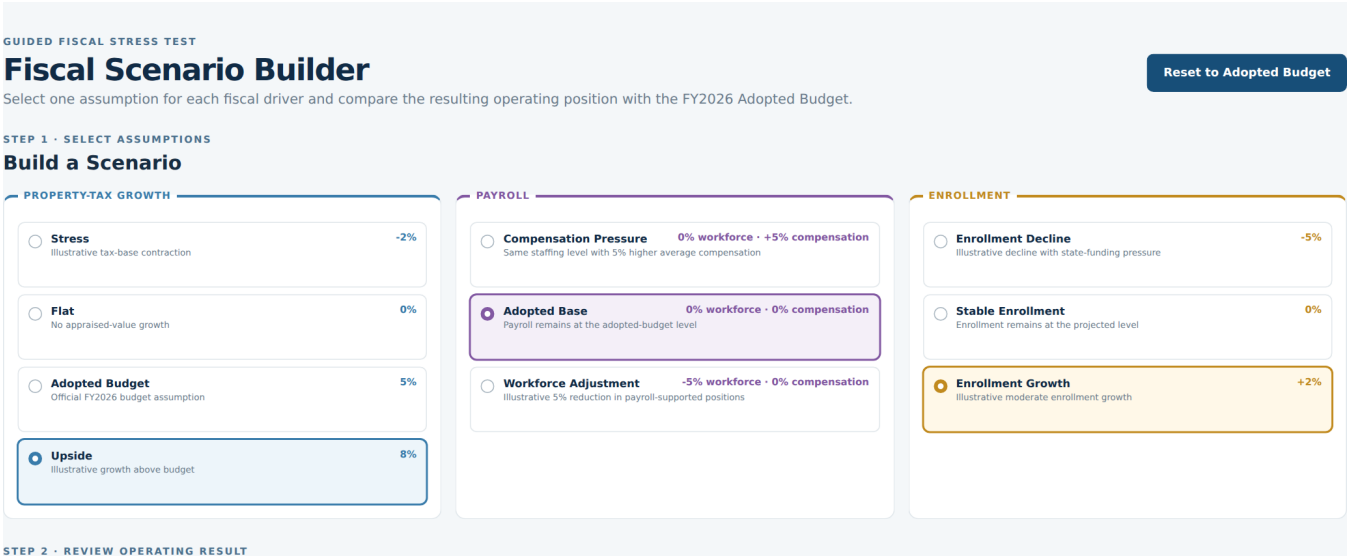


Figure 2. Actual guided controls: 8% tax growth, unchanged payroll and 2% enrollment growth.

Driver	Guided alternatives
Property tax	−2%, 0%, 5% adopted configuration, or 8%.
Payroll	5% compensation growth; unchanged payroll; or 5% workforce reduction.
Enrollment	5% decline, stable enrollment, or 2% growth.

The 5% tax configuration reproduces adopted revenue through an inferred pre-growth base. Alternative menu settings are illustrative. Payroll and enrollment changes are relative to the imported starting values.

Separate the mechanisms

Enrollment affects state revenue through a simplified per-pupil proxy. It does not automatically reduce payroll or change property-tax revenue. These effects require separate assumptions.

03 / INSIDE THE DASHBOARD

See the financial impact

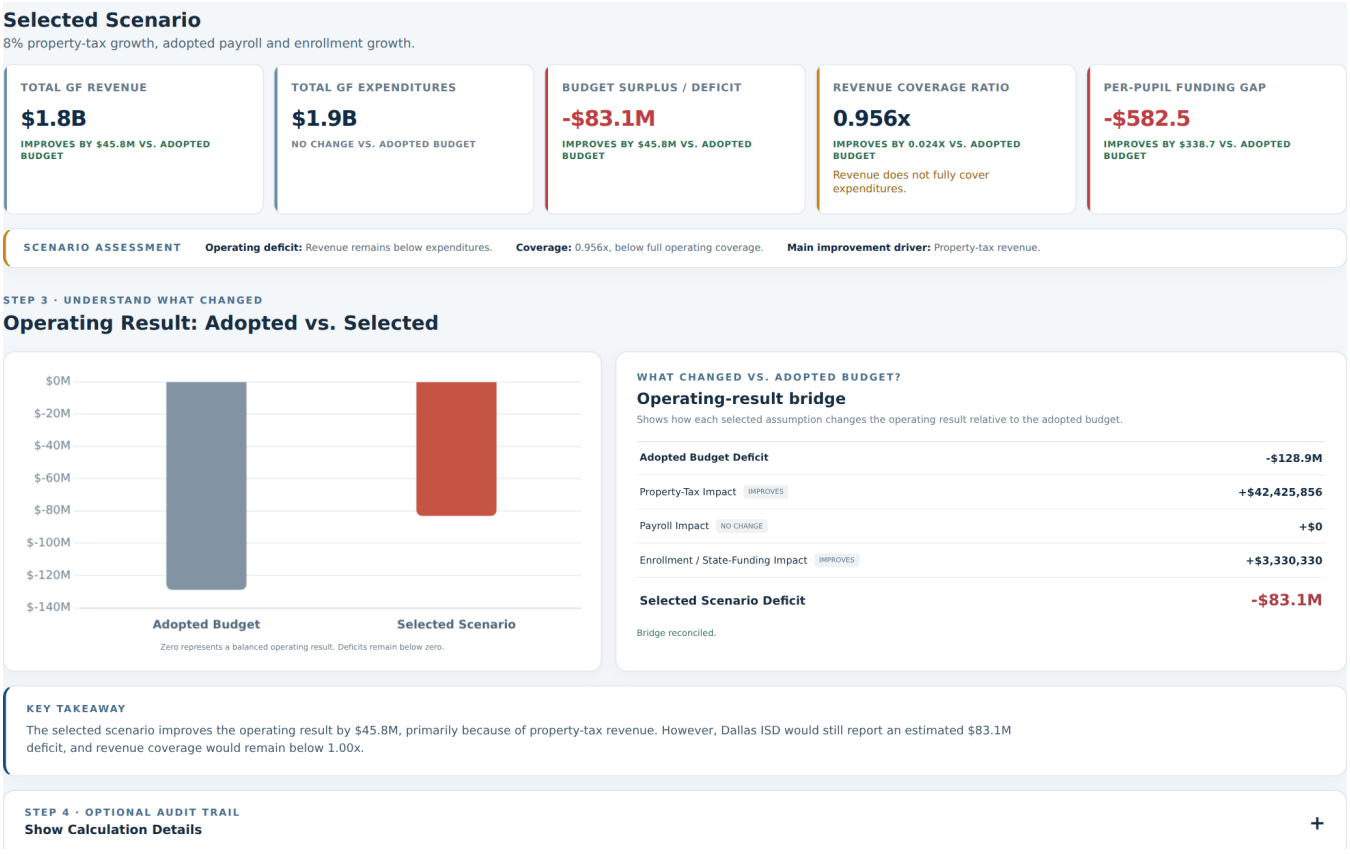


Figure 3. Actual scenario outputs and reconciled bridge. Capture taken after chart animation settled.

The upside combination improves the operating result by $45.8M but leaves an $83.1M deficit. Coverage rises to 0.956x, below full budgetary expenditure coverage.

Bridge component	Impact
Property-tax revenue	+$42,425,856
Enrollment / state revenue	+$3,330,330
Payroll	$0
Operating-result improvement	+$45,756,187

Displayed whole-dollar impacts can differ by $1 when individually rounded. The engine reconciles using unrounded values.

03 / INSIDE THE DASHBOARD

Identify the material risk drivers

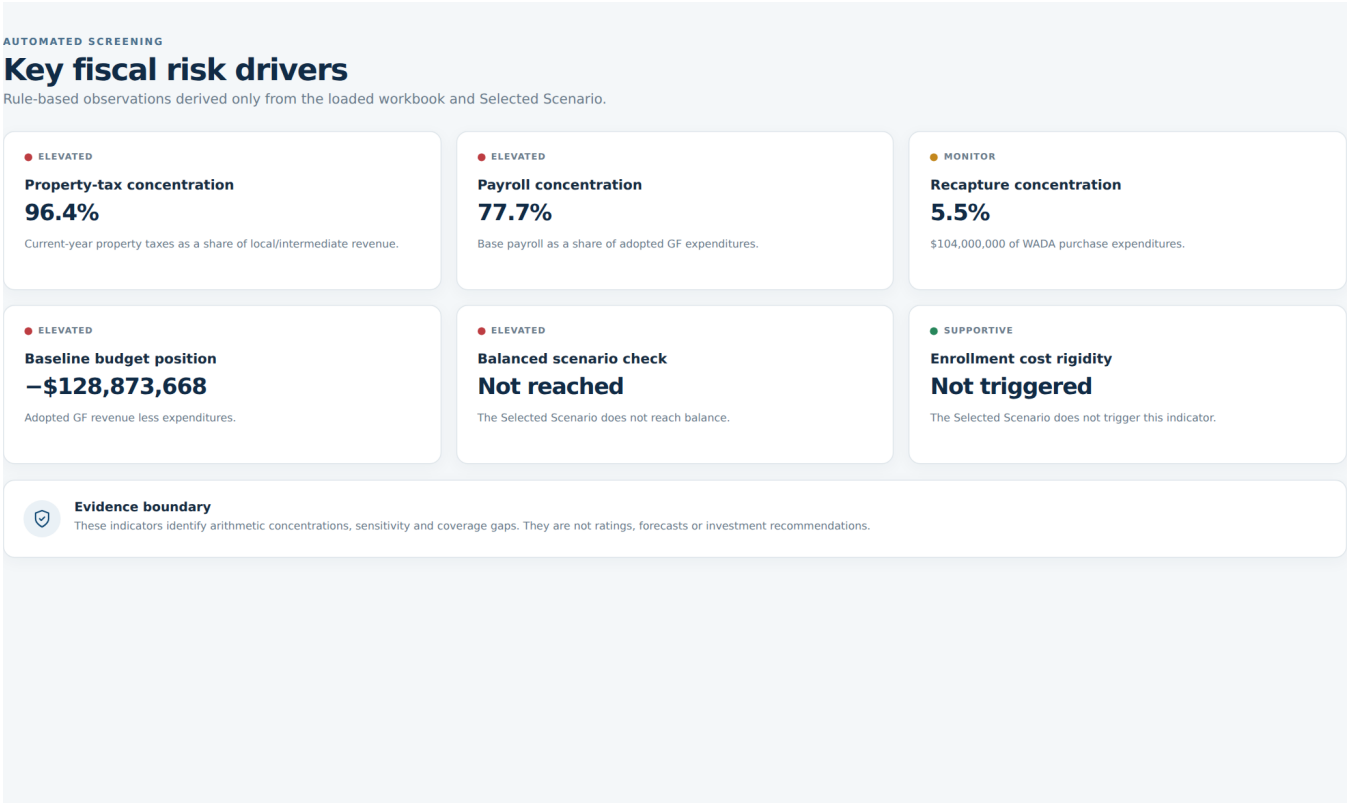


Figure 4. Actual Risk Drivers view, showing concentrations and enrollment exposure.

The view identifies property-tax and payroll concentration, recapture context and enrollment sensitivity. These measures inform the scenario selection; they are not a rating score.

Payroll costs are 77.7% of total expenditures. Excluding the $104M recapture amount, the official source describes payroll at 82.3% of the remaining expenditure base. The two ratios use different denominators. [1, p. 35]

Financial context is not a causal forecast

A large budget share makes an assumption financially material. It does not independently quantify the likelihood of a shock or the district’s ability to respond.

03 / INSIDE THE DASHBOARD

Inspect mathematical break-even conditions

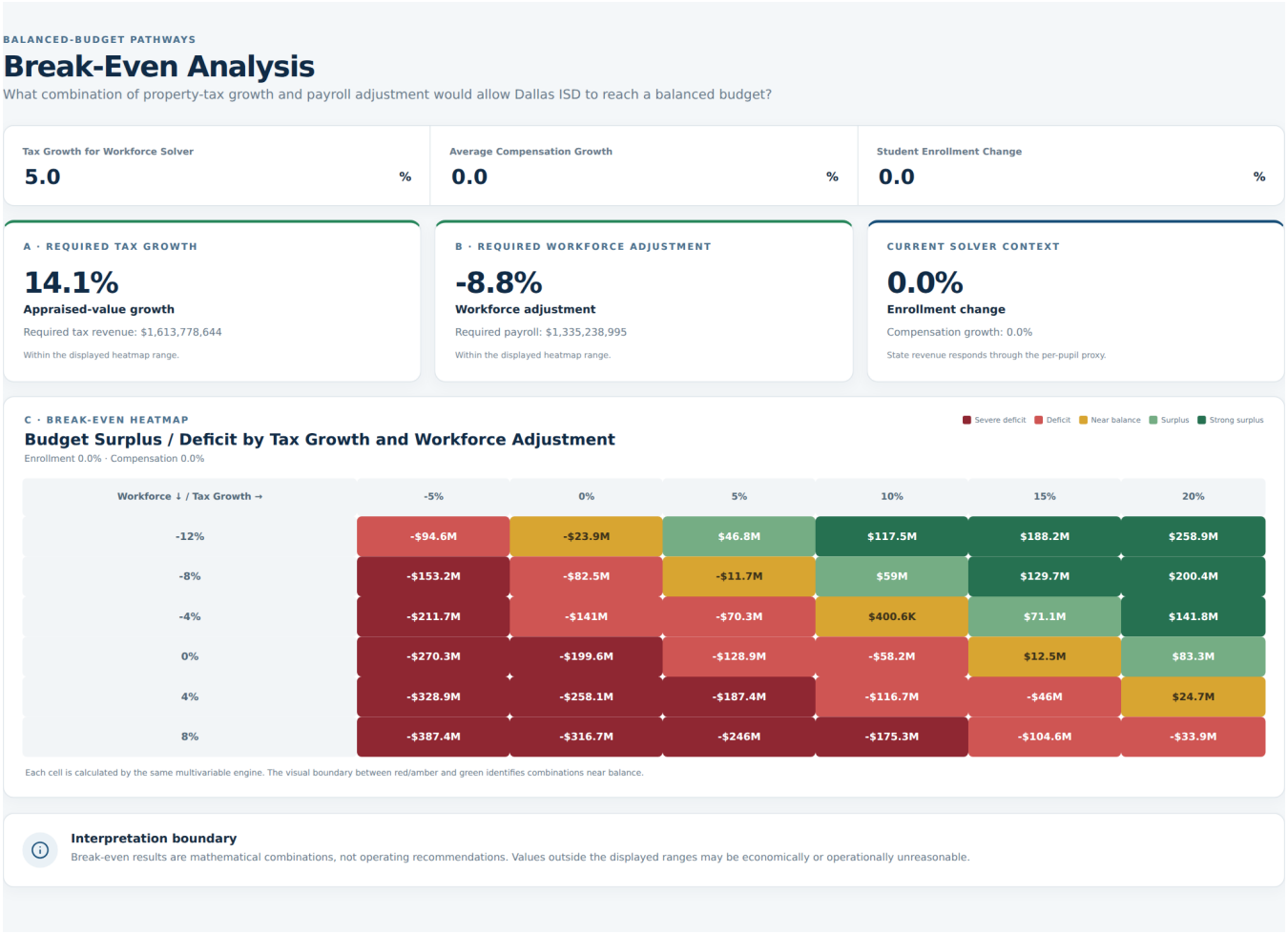


Figure 5. Actual Break-Even view at adopted revenue, enrollment and compensation settings.

The heatmap recalculates operating results across tax and workforce combinations, holding the selected compensation and enrollment assumptions constant. Its colors indicate a mathematical boundary rather than operational feasibility.

The algebraic solutions are approximately 14.11% tax growth relative to the inferred pre-growth base, or −8.80% workforce adjustment at adopted revenue. They are mathematical thresholds, not recommendations or assessments of feasibility.

03 / INSIDE THE DASHBOARD

Explore the path to recovery

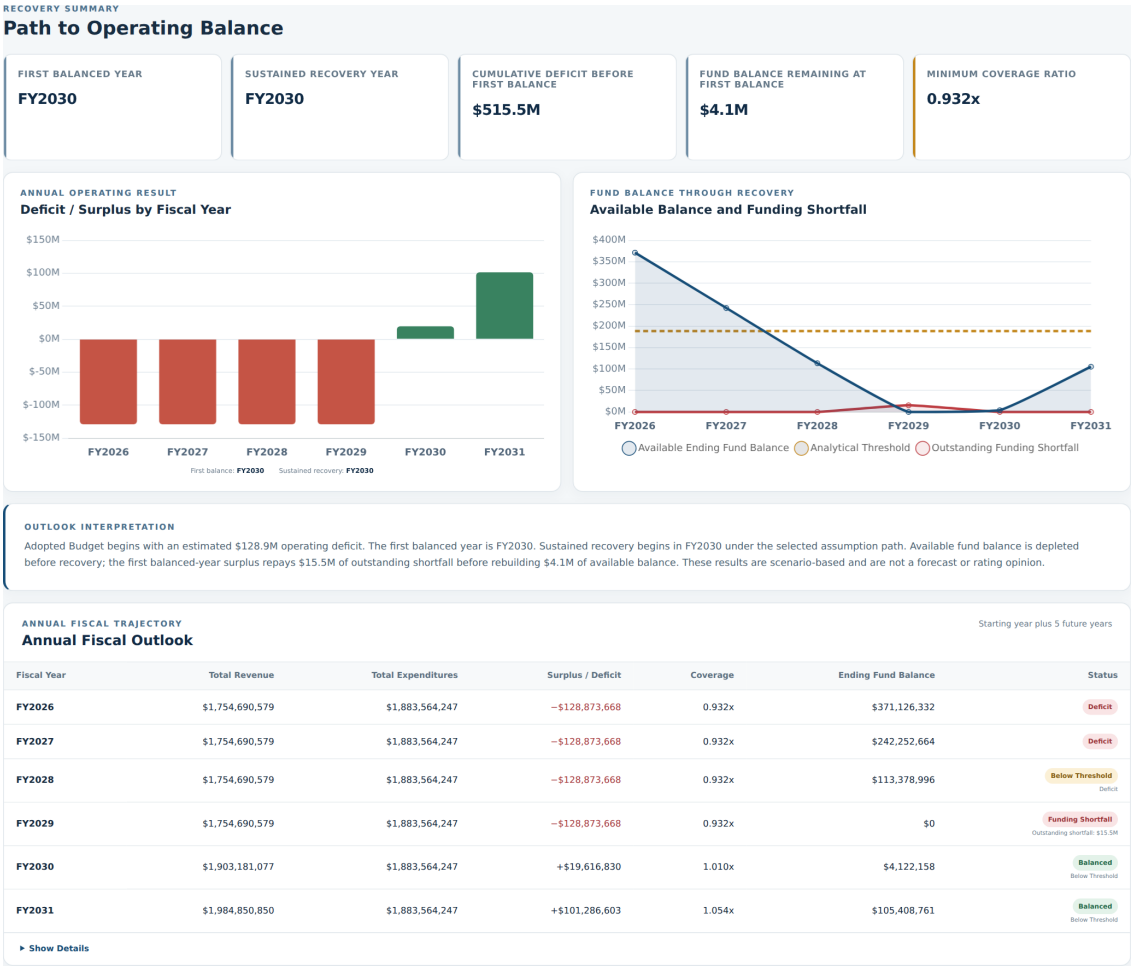


Figure 6. Actual recovery results for the illustrative FY2030 path; source type is Illustrative Assumption.

The example reaches first balance and two-year sustained recovery in FY2030. The assumed $500M beginning balance is exhausted before recovery. FY2030 surplus first repays $15.5M of shortfall, leaving $4.1M available.

A conditional result

The $500M reserve is not an official Dallas ISD balance. Tax growth is 0% in FY2027–FY2029, 10% in FY2030 and 5% in FY2031; future payroll and enrollment changes are 0%.

04 / FINANCIAL SOURCES & DATA PREPARATION

Source evidence and fiscal periods

The FY2025–2026 Adopted Budget Book is the principal source for the audited inputs. Page numbers below refer to the PDF page labels, which match the physical page numbers in the 609-page file. [1]

Source / page	What was verified
Budget Book, p. 25	Fund-level revenue and expenditures; General Operating cross-check.
Budget Book, p. 31	FY2025/FY2026 revenue, current taxes and local/state/federal totals.
Budget Book, p. 35	Payroll, total expenditures and $104M recapture footnote.
Budget Book, p. 83	139,890 projected students and 139,802 prior PEIMS enrollment.
Budget Book, pp. 16–17	FY2024 actual comparisons reproduced by the district with ACFR attribution.
Budget Book, p. 20	Budget-summary context and proposed future amendments.

The workbook labels its FY2026 columns “Adopted.” Some underlying tables in the published adopted book retain “Recommended Budget” or “Proposed Budget” headings. The report retains that source context rather than silently changing the original tables.

Supporting documents are not interchangeable

The community presentation is an earlier planning snapshot: its p. 9 total revenue is $1,758,690,579, $4M above the final workbook baseline. The final revenue table removes the $4M prior-year-tax budget shown for FY2025. The project uses the final book/workbook values rather than mixing versions. [1, p. 31; 3, p. 9]

FY2024 actual figures shown in the book do not establish an independently executed ACFR extraction. The principal implemented analysis remains adopted-budget analysis.

Provenance discipline

Identify the document version, fiscal period, fund and measure before comparing a number. Internal reconciliation confirms consistency, not source authenticity.

04 / FINANCIAL SOURCES & DATA PREPARATION

From the budget table to Excel

General Fund
Revenue Budget Comparison by Object
2025-2026

Object	Description	2025 Adopted Budget	2026 Recommended Budget	Amount Increase (Decrease)	% Incr (Decr)
5711	Taxes-Current Year	$ 1,352,676,224	$ 1,484,904,976	$ 132,228,752	9.8%
5712	Taxes-Prior Year	4,000,000	-	(4,000,000)	(100.0%)
5719	Penalties & Interest	10,000,000	10,000,000	-	0.0%
5739	Tuition And Fees From Local So	500,000	500,000	-	0.0%
5742	Invst Earnings	44,000,000	40,000,000	(4,000,000)	(9.1%)
5743	Rent Income	1,100,000	1,100,000	-	0.0%
5744	Gifts & Bequests	20,000	20,000	-	0.0%
5749	Other Revs from Loc Sources	3,000,000	3,000,000	-	0.0%
5752	Athletics Gate Revenue	1,000,000	1,000,000	-	0.0%
Total 5700 All Loc/Intermediate Rev		**$ 1,416,296,224**	**$ 1,540,524,976**	**$ 124,228,752**	**8.8%**
5811	Per Capita Apportionment	$ 48,893,924	$ 49,962,838	$ 1,068,914	2.2%
5812	Foundation-Sal & Op	93,504,190	32,553,673	(60,950,517)	(65.2%)
5831	TRS On-Behalf	80,000,000	84,000,000	4,000,000	5.0%
Total 5800 All State Prg Revs		**$ 222,398,114**	**$ 166,516,511**	**$ (55,881,603)**	**(25.1%)**
5919	Revenue From Federal Sources	$ 5,000,000	$ 5,000,000	$ -	0.0%
5928	Indirect Cost Through TEA	10,000,000	10,000,000	-	0.0%
5931	Sch Health & Related Svcs	15,000,000	6,849,092	(8,150,908)	(54.3%)
5939	Fed Rev By State Other Than TE	800,000	800,000	-	0.0%
5949	Misc Fed Rev	25,000,000	25,000,000	-	0.0%
Total 5900 All Fed Prg Revs		**$ 55,800,000**	**$ 47,649,092**	**$ (8,150,908)**	**(14.6%)**
	Total	**$ 1,694,494,338**	**$ 1,754,690,579**	**$ 60,196,241**	**3.6%**

Figure 7. Original Dallas ISD revenue table, cropped from Budget Book p. 31. Source headings are preserved.

Corresponding workbook ranges

GF Revenue range / label	FY25	FY26
D6 / Taxes - Current Year	$1,352,676,224	$1,484,904,976
D15 / Total Local/Intermediate Revenue	$1,416,296,224	$1,540,524,976
D21 / Total State Program Revenue	$222,398,114	$166,516,511
D29 / Total Federal Program Revenue	$55,800,000	$47,649,092
D31 / TOTAL GENERAL FUND REVENUE	$1,694,494,338	$1,754,690,579

This is a faithful transcription view of selected cached Excel values, not an Excel application screenshot. The D-column references document this workbook; the dashboard itself maps by labels and headers rather than these addresses.

The paired source and workbook view demonstrates traceability of the resulting data. It does not prove which extraction steps were automated.

04 / FINANCIAL SOURCES & DATA PREPARATION

Preparation record and workbook structure

The preparation history includes manual work and reported Python-assisted PDF extraction using pdfplumber. The original script and intermediate extraction files are not supplied. The resulting workbook is inspectable; the complete historical extraction process is not reproducible from this repository alone.

The supplied inspect_workbook.py is a read-only OOXML inspection utility using Python's standard library. It is not the original PDF extraction pipeline. No pandas, NumPy or statistical model is assumed to be part of the application.

Worksheet	Purpose
Budget Summary	Fund-level overview and expenditure cross-check.
GF Revenue	Tax, local, state and federal revenue.
GF Expenditures	Payroll, recapture and spending categories.
Other Funds	Food Service and Debt Service budget context.
Enrollment	Prior enrollment and FY2026 projected enrollment.
Scenario Analysis	Original tax-only spreadsheet reference scenarios.
Financial Ratios	Reference financial indicators and formulas.

The workbook contains seven worksheets, 358 formula elements, five charts and five comments, with zero structured Excel table parts. The dashboard does not execute those spreadsheet formulas; it calculates its own outputs from mapped inputs.

Original Excel scenarios versus current application

The spreadsheet retains tax-only reference cases. The dashboard expands the starting scenario to include payroll and enrollment, then adds the separate annual outlook. These are related stages, not identical implementations.

04 / FINANCIAL SOURCES & DATA PREPARATION

Mapping, cleaning and reconciliation

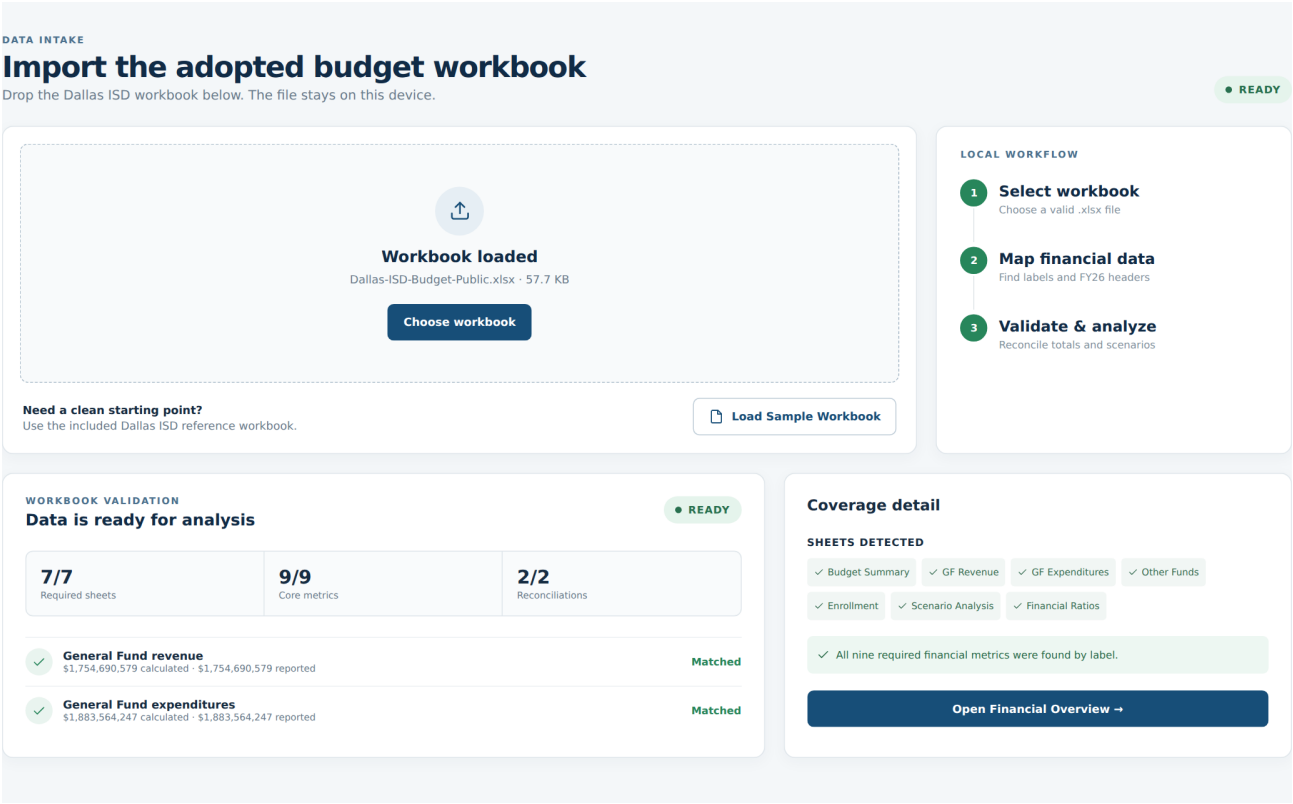


Figure 8. Actual local intake and validation view for the public sample workbook.

Mapping normalizes case, accents, punctuation, whitespace and line breaks. The nearest matching fiscal-year header above a financial label identifies the value column. Numeric text handling removes currency symbols and commas and interprets parentheses as a negative sign.

> Local + State + Federal revenue = General Fund revenue
> $1,540,524,976 + $166,516,511 + $47,649,092 = $1,754,690,579

The expenditure total is compared with General Operating expenditures in Budget Summary. Both checks use a $1 tolerance. Ready means required information and reconciliations pass; Warning permits analysis with a reconciliation difference; Error blocks analytical views when required data is missing.

Public-workbook preparation changed only workbook metadata and document properties. Financial cells, formulas, cached values, charts and comments were preserved. The embedded sample is this sanitized public copy.

05 / FINANCIAL MODELING METHODOLOGY

Baseline and property-tax sensitivity

The model is deterministic: identical inputs and assumptions produce identical outcomes. It does not assign scenario probabilities.

> Operating result = Total revenue − Total expenditures
> Revenue coverage ratio = Total revenue / Total expenditures

At baseline: $1,754,690,579 − $1,883,564,247 = −$128,873,668. Coverage is 0.93157989264x. Revenue covers about 93.2% of budgeted expenditures.

Replace the growth assumption; do not add it twice

> Implied pre-growth tax revenue = Adopted tax revenue / 1.05
> = $1,414,195,215.24
>
> Scenario tax revenue = Implied pre-growth tax revenue × (1 + selected growth)

Selecting 5% reproduces adopted tax revenue. Selecting 8% applies 8% to the inferred base; it does not increase adopted tax revenue by another 8%.

Important denominator distinction

The source table shows current-tax revenue increasing about 9.8% from FY2025 to FY2026. The engine's 5% adopted configuration is an explicit model calibration, not that reported year-over-year change. Its inferred base is not the FY2025 tax-revenue value. [1, p. 31]

The 5% taxable-appraised-value growth assumption is documented in the district's April 10, 2025 planning presentation, p. 5. The engine uses it as a simplified revenue-factor calibration. The model does not recreate the tax-rate and collection assumptions also shown on that slide. [4, p. 5]

Tax rates, exemptions, assessed-value mechanics and collection behavior are not separately modeled.

05 / FINANCIAL MODELING METHODOLOGY

Enrollment, payroll and financial outputs

Scenario enrollment = Starting enrollment × (1 + enrollment change)
Starting state revenue per pupil = Starting state revenue / Starting enrollment
Scenario state revenue = Starting state revenue per pupil × Scenario enrollment

The proxy produces proportional state-revenue changes. It does not recreate weighted attendance, the Texas funding formula or variable recapture. Enrollment does not automatically change staffing or property taxes.

Scenario payroll = Adopted payroll × (1 + workforce adjustment)
× (1 + compensation growth)
Scenario expenditures = Scenario payroll + Fixed non-payroll expenditures

Non-payroll expenditures are $419,451,584. Workforce and compensation effects multiply; the model does not specify positions, benefits, severance or implementation timing.

Scenario revenue = Tax + Other local + State + Federal
Revenue per pupil = Revenue / Enrollment
Expenditure per pupil = Expenditures / Enrollment
Per-pupil gap = Revenue per pupil − Expenditure per pupil

Other local revenue is fixed at $55,620,000 and federal revenue at $47,649,092. Safe division returns an unavailable result when the denominator is zero. Enrollment precision is retained internally and rounded for display.

Interpret per-pupil changes carefully

A lower student count can increase expenditure per pupil when spending does not adjust. This is a mechanical ratio effect, not evidence that service levels or financial resilience improved.

05 / FINANCIAL MODELING METHODOLOGY

Attribution and mathematical break-even

Tax impact = Scenario tax − Adopted tax
Enrollment/state impact = Scenario state − Adopted state
Payroll impact = −(Scenario payroll − Adopted payroll)

Scenario result = Adopted result + Tax impact + State impact + Payroll impact

Other local, federal and non-payroll categories are fixed, so the three impacts explain the modeled change. Positive impacts improve the result; negative impacts worsen it. The bridge reconciles within $1.

Property-tax solver

Required tax = Scenario expenditures − Other local − Scenario state − Federal
Required growth = Required tax / Implied pre-growth tax − 1

Workforce solver

Required payroll = Scenario revenue − Non-payroll expenditures
Required workforce change = Required payroll /
[Adopted payroll × (1 + compensation growth)] − 1

At adopted assumptions, the results are 14.11% tax growth and −8.80% workforce adjustment. Tax growth is measured relative to the inferred base, about 9.1 percentage points above the model's adopted 5% setting.

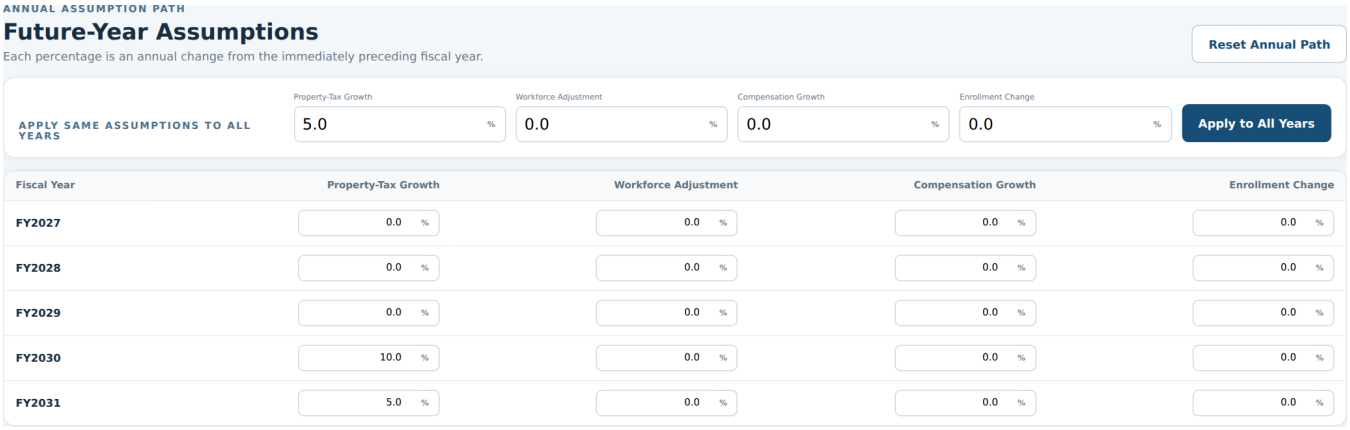
The sensitivity heatmap evaluates tax growth from −5% to +20% and workforce adjustments from −12% to +8%, with enrollment and compensation settings held at the selected values.

Scope of the solver

Algebraic balance does not establish economic likelihood, legal permissibility or operational feasibility. The solutions frame additional research rather than prescribe action.

05 / FINANCIAL MODELING METHODOLOGY

Annual paths and recovery definitions


ANNUAL ASSUMPTION PATH

Future-Year Assumptions

Each percentage is an annual change from the immediately preceding fiscal year.

Reset Annual Path

APPLY SAME ASSUMPTIONS TO ALL YEARS

Property-Tax Growth	Workforce Adjustment	Compensation Growth	Enrollment Change	
5.0 %	0.0 %	0.0 %	0.0 %	Apply to All Years

Fiscal Year	Property-Tax Growth	Workforce Adjustment	Compensation Growth	Enrollment Change
FY2027	0.0 %	0.0 %	0.0 %	0.0 %
FY2028	0.0 %	0.0 %	0.0 %	0.0 %
FY2029	0.0 %	0.0 %	0.0 %	0.0 %
FY2030	10.0 %	0.0 %	0.0 %	0.0 %
FY2031	5.0 %	0.0 %	0.0 %	0.0 %

Figure 9. Actual annual inputs used in the FY2030 illustrative example.

$$\text{Tax}(t) = \text{Tax}(t-1) \times [1 + \text{tax growth}(t)]$$
$$\text{Enrollment}(t) = \text{Enrollment}(t-1) \times [1 + \text{enrollment change}(t)]$$
$$\text{Payroll}(t) = \text{Payroll}(t-1) \times [1 + \text{workforce change}(t)] \times [1 + \text{compensation growth}(t)]$$
$$\text{State}(t) = \text{Starting state revenue per pupil} \times \text{Enrollment}(t)$$

Other local revenue, federal revenue and non-payroll expenditures remain at their starting-point amounts. Horizons contain the starting year plus three, five or seven future years. Default future tax growth is 5%, with other annual changes at 0%; those defaults are illustrative.

First Balanced Year: first non-negative operating result. Sustained Recovery Year: first of two consecutive non-negative results with coverage ≥1.00x. A positive final year alone cannot confirm sustained recovery.

Starting assumptions versus future changes

An 8% starting tax scenario replaces the 5% setting relative to the inferred base. A subsequent 5% annual assumption increases that resulting tax revenue by 5% in the next year. The starting scenario is not reapplied each year.

05 / FINANCIAL MODELING METHODOLOGY

Reserve reconciliation and funding shortfalls

Net position = Beginning available balance − Beginning shortfall
+ Annual operating result

Ending available balance = MAX(0, Net position)
Ending shortfall = MAX(0, −Net position)

Both ending positions roll into the following year. They cannot both be positive. A later surplus repays the outstanding shortfall before rebuilding reserves. The shortfall is a modeled funding requirement, not automatically debt or an accounting liability.

FY2030 reconciliation	Amount
Beginning available balance	$0
Beginning funding shortfall	$15,494,672
Operating surplus	$19,616,829.60
Ending funding shortfall	$0
Ending available balance	$4,122,157.60

Available balance ratio = Ending available balance / Annual expenditures
Threshold amount = Annual expenditures × Selected threshold percentage

The beginning balance is optional and applies at the beginning of the starting fiscal year. Blank inputs do not block the operating outlook. Reserve metrics then remain unavailable.

Analytical reserve assumption

The example uses $500M; no official Dallas ISD reserve value is imported or hardcoded. The default 10% threshold is not district policy or a rating standard. Fund balance does not necessarily equal cash liquidity.

06 / MODEL VALIDATION & TESTING

What the tests establish

Verification addresses input completeness, calculation accuracy and interface consistency. It does not establish the probability of a scenario or the economic validity of an assumption path.

Area	Checks
Workbook and baseline	Labels and headers; required inputs; adopted-budget parity; revenue and expenditure reconciliation.
Scenarios and attribution	Revenue/payroll/enrollment formulas; combined effects; 36 guided combinations reconcile in the delivery record.
Break-even	Algebraic solutions produce near-zero operating results.
Annual outlook	Compounding; fixed categories; selected starting scenario; recovery detection.
Fund balance	Optional input; depletion floor; outstanding-shortfall repayment; mutual exclusivity.

The supplied suite contains 30 unit tests: 16 workbook/scenario tests and 14 outlook tests. Phase 1A records all passing after clean installation and dependency updates. These are recorded local verification results; the documentation workflow did not repeat the entire unit suite.

A targeted independent numerical check

For this report, the supplied JavaScript engine was run directly against the nine retained inputs. The upside result was −$83,117,481.32. The recovery path returned FY2030 and $4,122,157.60 ending available balance, matching the reference case.

Three distinct recovery conditions

• No balanced year within the horizon.

• First balance only in the final year: sustained recovery unconfirmed.

• Two consecutive balanced years: sustained recovery begins in the first of those years.

06 / MODEL VALIDATION & TESTING

Browser, integrity and distribution checks

Phase 1A records full Chrome smoke workflows through both local HTTP preview and file://, including original-workbook upload, scenario controls, break-even analysis and multi-year conditions.

For the report, a fresh headless Chrome session opened the final standalone HTML, loaded the public sample, visited the analytical tabs and entered the illustrative FY2030 path. The captures show settled charts; no page errors or external HTTP requests were observed during this session.

Evidence	Scope
30 unit tests	Passing result recorded in Phase 1A; not re-executed for report layout.
HTTP and file workflows	Full smoke-test results recorded in Phase 1A.
New report capture session	Standalone sample loading and selected scenario/outlook path observed independently.
Responsive review	Phase 1A records no page overflow at 1440, 1280 and 390 px widths.
Workbook integrity	Only xl/workbook.xml and docProps/core.xml changed from the protected source.
Financial engine integrity	Six calculation/intake/helper modules byte-identical to the prior audited copy.

The public workbook has seven sheets, 358 formula elements, five charts and five comments. Its SHA-256 matches the sample embedded in dist/index.html. Exact fingerprints appear in the appendix.

Implementation verification versus economic validation

A correctly implemented formula can still simplify reality. Economic validation would require more historical evidence, funding-system detail, forecast calibration and management feasibility analysis.

07 / KEY FINDINGS & RESEARCH IMPLICATIONS

The budget improves but remains below balance

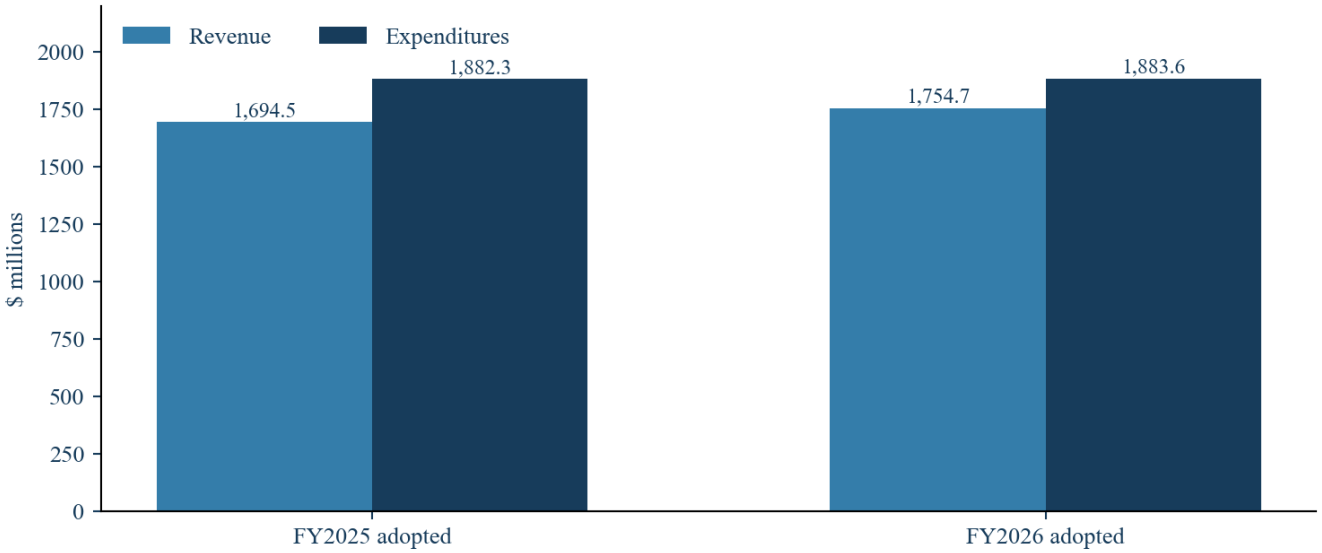


Figure 10. Calculated from FY2025/FY2026 budget inputs; dollars in millions. [1, pp. 31, 35]

Measure	FY2025	FY2026
Revenue	$1,694.5M	$1,754.7M
Expenditures	$1,882.3M	$1,883.6M
Operating result	−$187.8M	−$128.9M
Coverage	0.900x	0.932x

The budgeted gap narrows by $58.9M. Revenue rises $60.2M while expenditures rise $1.2M. This is a comparison of adopted planning amounts, not a measured improvement in actual operating performance.

Follow-up credit research

How did actual results compare with adopted budgets? Which reserves, amendments or management actions supported the planned gap? Are the revenue changes recurring?

07 / KEY FINDINGS & RESEARCH IMPLICATIONS

Material drivers and combined pressure

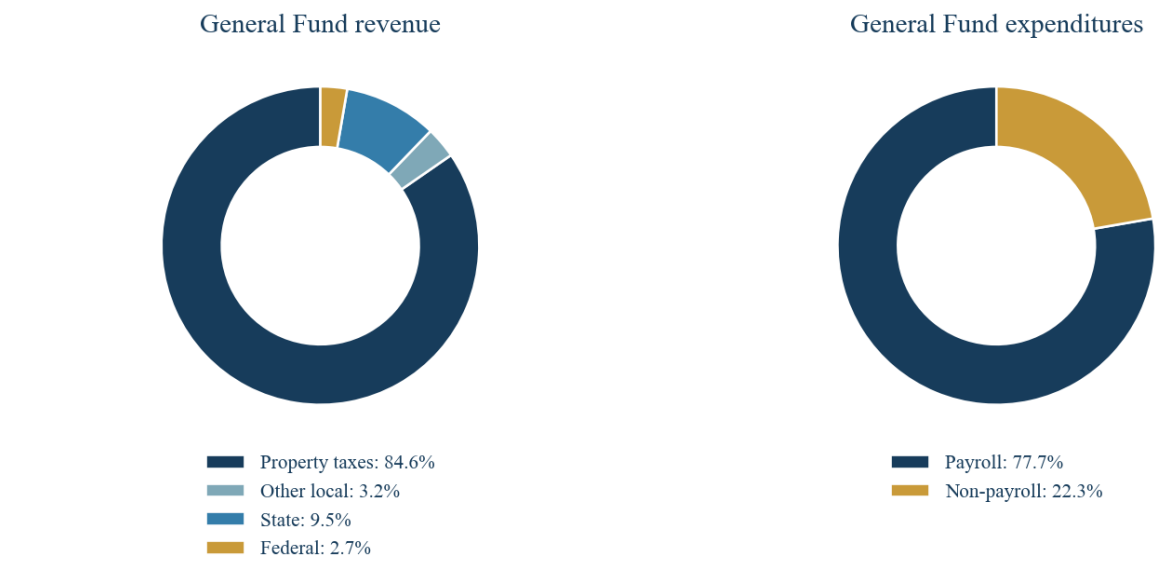


Figure 11. FY2026 budget composition. Taxes are 84.6% of total GF revenue and 96.4% of local revenue; payroll is 77.7% of expenditures.

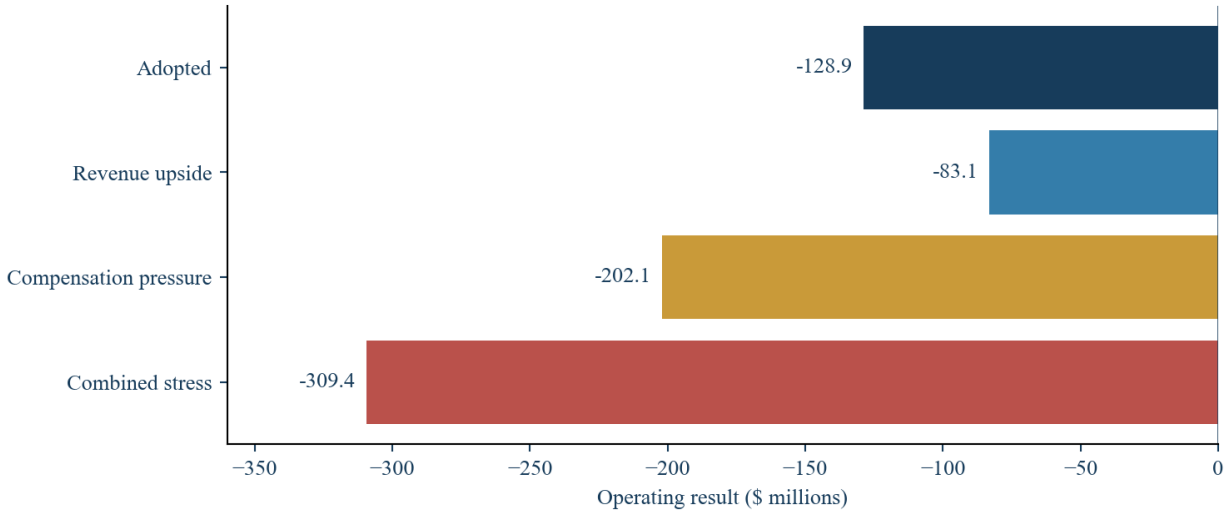


Figure 12. Engine-calculated operating results. Alternatives are illustrative, with assumptions listed below.

Upside: tax 8%, enrollment +2%, unchanged payroll. Compensation: tax 5%, compensation +5%, other changes 0%. Combined stress: tax −2%, compensation +5%, enrollment −5%, workforce unchanged.

The upside improves the deficit without closing it. The combined stress produces a $309.4M deficit. These calculations motivate broader analysis of recurring gaps, tax-base resilience and spending flexibility; they do not prove a permanent structural deficit or assign scenario probabilities.

07 / KEY FINDINGS & RESEARCH IMPLICATIONS

Recovery timing and reserve capacity differ

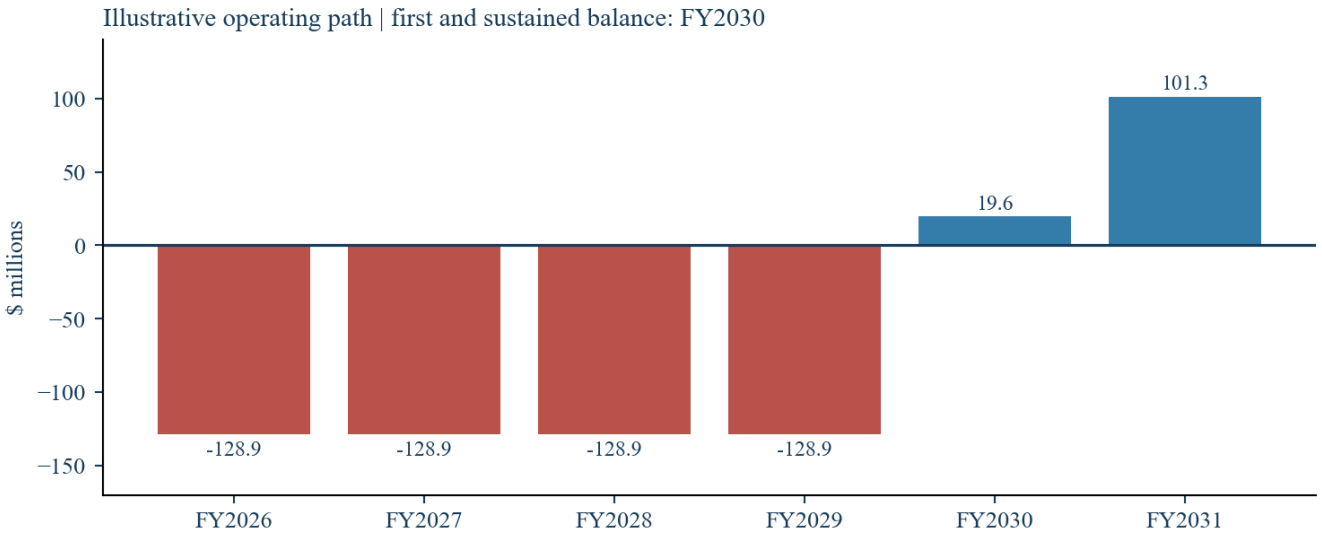


Figure 13. Illustrative operating trajectory generated by the supplied engine.

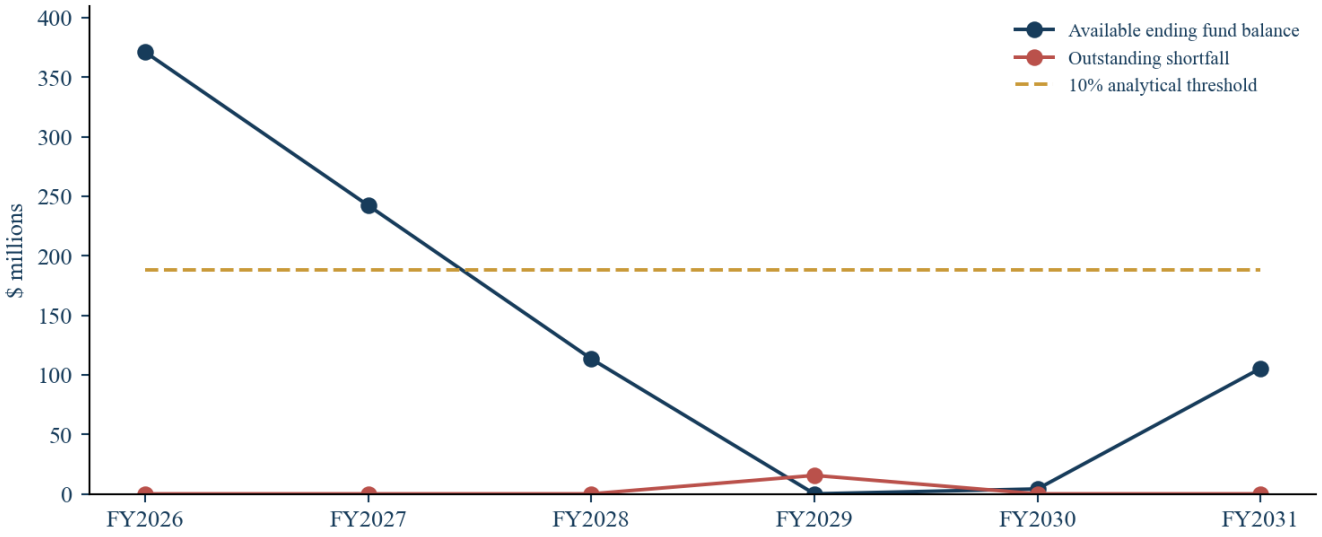


Figure 14. $500M illustrative beginning balance; available reserves and outstanding shortfall remain non-negative and mutually exclusive.

Four deficits from FY2026 through FY2029 total $515.5M. The $500M assumption is exhausted before operating balance; the resulting $15.5M shortfall is repaid in FY2030. Reserve rebuilding is only $4.1M that year.

Research implication

A balanced terminal year is insufficient to assess the transition. Reserve availability, restrictions, intra-year liquidity and financing responses require separate investigation.

08 / ASSUMPTIONS & LIMITATIONS

Interpret the model within its boundaries

Boundary	Interpretation
Budget versus actual	Adopted amounts are plans. A budgeted deficit is not a realized loss or an automatic reduction in reported reserves.
Property taxes	Revenue-factor sensitivity; no separate rates, exemptions, collections or valuation model. The 5% engine calibration is not the source-table 9.8% revenue comparison.
State funding	Fixed per-pupil proxy; no full Texas funding, weighted attendance or dynamic recapture.
Cost structure	Proportional payroll factors; non-payroll fixed. No inflation, position-level staffing or implementation expense.
User-defined paths	Source types are user-classified. No independent forecast validation or probability estimates.
Fund balance	Optional user input; no restriction classification or cash-flow timing. Shortfall is an analytical funding requirement.
Recovery	Two-year confirmation is an application convention, not permanent sustainability or a rating criterion.
Credit scope	No comprehensive debt, pensions, economic-base, governance, legal-security or institutional-support assessment.
Evidence	Original extraction script unavailable; source-to-workbook result is inspectable.

The tool supports research and conditional analysis. It does not assign credit ratings, estimate default probability, forecast official outcomes or recommend investments.

09 / TECHNICAL IMPLEMENTATION & REPRODUCIBILITY

Inspect, run and reproduce

Component	Verified implementation
Frontend	HTML, CSS and vanilla JavaScript; src/main.js and src/styles.css.
Intake	workbook.js, mapping.js, validation.js; SheetJS 0.20.3.
Financial logic	calculations.js for scenarios/attribution/solvers; runway.js for current Multi-Year Outlook.
Presentation	Chart.js 4.5.1 and format.js.
Build	Vite 6.4.3; vite-plugin-singlefile 2.3.3.
Verification	Vitest 4.1.11; Playwright Core 1.62.1.
Python	Standard-library workbook inspection utility; original pdfplumber extraction not supplied.

```
npm ci
npm test
npm run dev
npm run build
```

Use Node.js 22 or later; Phase 1A was verified with Node 24.12.0. The dev URL is normally http://127.0.0.1:5173/. The final artifact is dist/index.html and can open directly as a local file. Installation needs dependency-network access; runtime workbook processing stays local.

Browser tests require Chrome, a configured executable path where necessary and a local preview for HTTP checks. No database, backend, authentication or statistical forecasting engine is implemented.

Codex supported coding, verification and technical hardening. Research framing, assumptions, formula review and financial interpretation remain the author’s responsibility.

The dated Phase 1A audit reported zero production vulnerabilities and three high entries in the development/build braces → micromatch → singlefile chain. This is a recorded audit result, not a security guarantee.

10 / CONCLUSIONS & POTENTIAL EXTENSIONS

A research question made testable

The project connects public-document research, an Excel source model and a transparent analytical application. It establishes a starting budget position, validates imported inputs and explains how explicit assumptions affect the fiscal outcome.

The central contribution

Users can move from a budget gap to an alternative scenario, understand the driver impacts and examine the timing and funding requirements of a selected recovery path.

The findings distinguish improvement from balance, and operating balance from reserve sufficiency. The model's simplicity makes these relationships inspectable, while its stated boundaries identify where deeper issuer research is required.

Potential extensions — not current features

Next research step	Purpose
Reproducible extraction	Recover or transparently reconstruct the original PDF-to-Excel pipeline.
Actuals and amendments	Compare plans with realized performance over a longer history.
State-funding detail	Reduce reliance on the proportional enrollment proxy.
Staffing and cost detail	Assess practical expenditure flexibility and adjustment costs.
Reserves and cash timing	Investigate usable balances, restrictions and liquidity.
External annual paths	Document and evaluate assumptions from separately developed models.

Dallas ISD is the case study. Extending the framework to another issuer requires re-examining data mapping, revenue mechanisms and institutional context.

Financial risk becomes easier to discuss when source information, assumptions and consequences can be traced. This application provides one practical implementation of that research approach.

11 / REFERENCES & PROJECT ACCESS

Sources and supporting evidence

1. Dallas ISD. FY2025–2026 Adopted Budget Book.

Open official source PDF

2. Dallas ISD. FY2025–2026 Budget Summary.

Open official source PDF

3. Dallas ISD. FY2025–2026 Community Budget Presentation.

Open official source PDF

4. Dallas ISD. April 10, 2025 Budget Planning Presentation.

Open official source PDF

Documents retrieved for report preparation on October 11, 2026. The principal numerical references are Budget Book pp. 25, 31, 35 and 83. Debt Service context is on p. 17. The 5% planning assumption is documented in reference 4, p. 5. Supporting presentation p. 9 is an earlier version and is not substituted for the final baseline.

Project evidence

• Dallas-ISD-Budget-Public.xlsx: sanitized financial source workbook.

• src/lib/: calculation, mapping and validation implementation.

• tests/: unit suite and browser smoke workflow.

• docs/WORKBOOK_ANALYSIS.md: workbook reference notes.

• README.md and STATUS.md: execution and delivery records.

• dist/index.html: audited standalone application.

Access status

GitHub, hosted-demo and release URLs have not been established at the time of this report. They will be added after author approval; no public availability is implied. The report’s official-source links remain usable.

Independent project

This is not an official Dallas ISD product, a credit-rating model, an agency rating opinion or an investment recommendation. All alternative scenarios and reserve examples are explicitly conditional.

APPENDIX A / FINANCIAL INPUT REGISTER

Trace each core input

Input / workbook location	Value	Budget Book
Current taxes / GF Revenue D6	$1,484,904,976	p. 31
Local revenue / GF Revenue D15	$1,540,524,976	p. 31
State revenue / GF Revenue D21	$166,516,511	p. 31
Federal revenue / GF Revenue D29	$47,649,092	p. 31
GF revenue / GF Revenue D31	$1,754,690,579	p. 31
GF expenditures / GF Expenditures E24	$1,883,564,247	p. 35
Payroll / GF Expenditures D28	$1,464,112,663	p. 35
Recapture / GF Expenditures E21	$104,000,000	p. 35 footnote
Enrollment / Enrollment C8	139,890	p. 83

General Operating expenditures on p. 25 provide the additional cross-check. Recapture is included within total expenditures and is not added twice or independently varied. Taxes are within local revenue, and payroll is within expenditures.

Distribution fingerprints

Public workbook SHA-256:
c46766264491fa250a70497e9738974dfe50127d592207cf2b427a90f1f8b7a7

Standalone HTML SHA-256:
8b99923a43b970151c4102dd2fad1cc6b07292f8bbb4069eaebfff68e44305fc

The HTML embeds a workbook with the same public-workbook fingerprint. The source workbook has 58 internal ZIP members; only the metadata parts changed in preparing the public copy.

Companion deliverables

This full report is separate from the planned three-page LinkedIn showcase. Public repository and hosting links will be added after those publication decisions are approved.

APPENDIX B / ADDITIONAL SOURCE EVIDENCE

Payroll, recapture and enrollment

General Fund
Expenditure Budget Comparison by Object Total
2025-2026

Object	Description	2025 Adopted Budget	2026 Recommended Budget	Amount Increase (Decrease)	% Incr (Decr)	Percent Budget
6100	Payroll Costs[1]	$ 1,454,794,349	$ 1,464,112,663	$ 9,318,314	0.6%	77.7%
6200	Prof/Contracted Svcs[2]	235,641,861	264,035,889	28,394,028	12.1%	14.0%
6300	Supplies/Materials	136,243,403	103,119,787	(33,123,616)	(24.3%)	5.5%
6400	Other Op Costs	36,673,773	35,157,708	(1,516,065)	(4.1%)	1.9%
6500	Debt Services	12,852,237	12,852,237	-	0.0%	0.7%
6600	Capital Outlay	6,111,349	4,285,963	(1,825,386)	(29.9%)	0.2%
	Total	**$ 1,882,316,972**	**$ 1,883,564,247**	**$ 1,247,275**	**0.1%**	**100.0%**

1. 2026 Recommended Budget includes a payroll cost percentage of 82.3% if recapture is excluded
2. 2026 Recommended Budget includes $104,000,000 for recapture

Figure 15. Actual source excerpt, Budget Book p. 35; payroll, total expenditures and recapture footnotes.

TOTAL ENROLLMENT SUMMARY			
High School Total	40,987	39,548	(1439)
Middle School Total	28,787	27,875	(912)
Elementary School Total	70,028	72,467	2,439
TOTAL ALL CAMPUSES	139,802	139,890	88

Notes:
2025-2026 Projected Enrollment provided by the Demographic Department
2025-2026 Projected Enrollment with unadjusted Prek 3 Enrollment
Orgs. 096, 099, 299 and 302 do not carry General Operating funds.

Figure 16. Actual source excerpt, Budget Book p. 83; projected enrollment is 139,890.

Payroll equals $1,464,112,663 and total General Fund expenditures equal $1,883,564,247. The footnote identifies $104M recapture within contracted services. Enrollment is a projection, not average daily attendance.

APPENDIX B / ADDITIONAL SOURCE EVIDENCE

The planning assumption behind the calibration

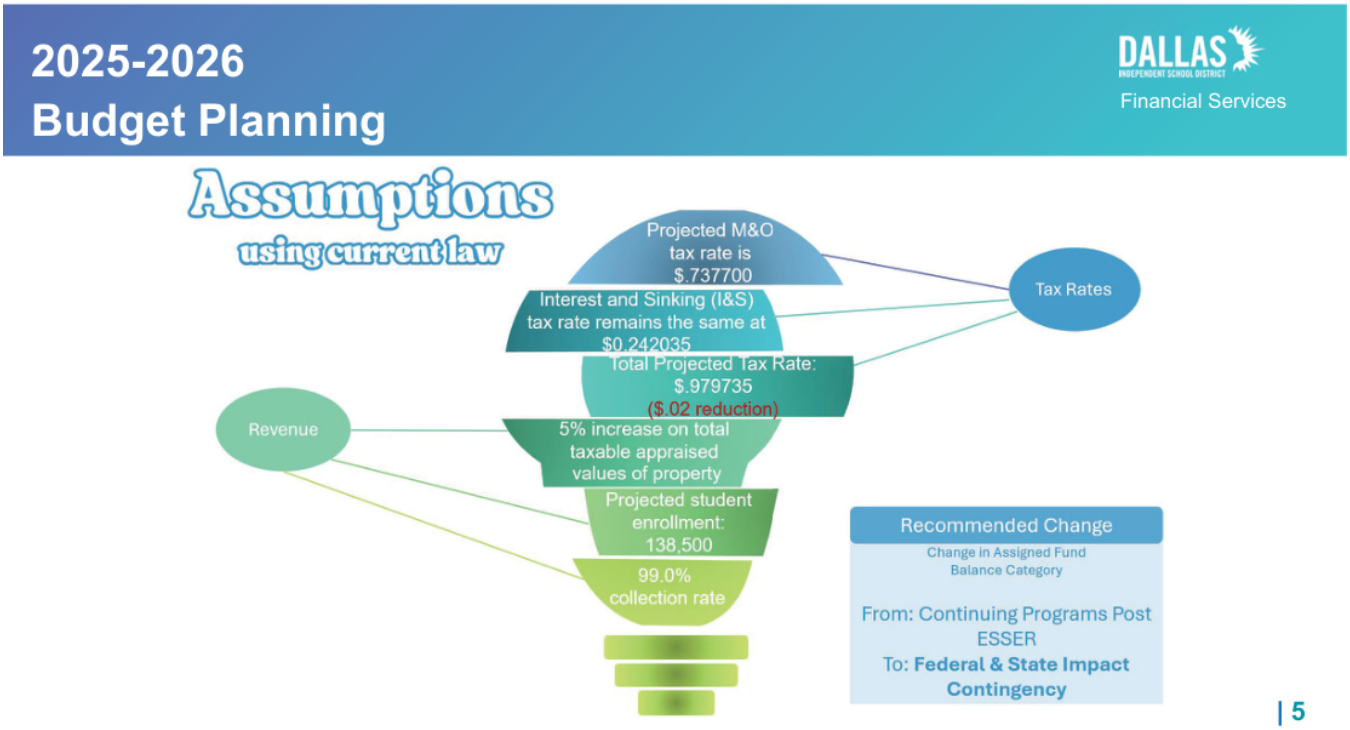


Figure 17. Actual April 10, 2025 planning slide, p. 5. The source documents 5% taxable-appraised-value growth. [4]

This earlier planning slide also contains tax-rate, collection and enrollment assumptions. It establishes the provenance of the 5% growth setting; its other values are not imported as the final baseline.

The final workbook uses 139,890 projected students from the adopted book, rather than the 138,500 shown on this earlier slide. This distinction prevents planning versions from being combined inadvertently.

A simplified sensitivity relationship

Dividing adopted tax revenue by 1.05 creates an inferred calibration base. It does not recover a historical assessed value or replicate the complete tax-collection calculation.